

# Microbiology Labs





## About the Course

Labs are an essential part of science in which students engage with the Things they are reading about and practice the scientific method.

This topic is included in the following course(s): Science: Microbiology



## Scheduling

| GRADE | SCHEDULE INFO.                          | BOOKS                                         |
|-------|-----------------------------------------|-----------------------------------------------|
| 12    | Microbiology Labs<br>1 time/week 60 min | Illustrated Guide to Home Biology Experiments |

### Sample Weekly View

| Day 1                                                        | Day 2                | Day 3                | Day 4                                                | Day 5                                     |
|--------------------------------------------------------------|----------------------|----------------------|------------------------------------------------------|-------------------------------------------|
| <b>Science: Microbiology</b>                                 |                      |                      |                                                      |                                           |
| Microbiology Lessons<br>Nature Walks & Scouting: Grades 9-12 | Microbiology Lessons | Microbiology Lessons | Microbiology Lessons<br>Nature Notebook: Grades 9-12 | Microbiology Lessons<br>Microbiology Labs |



## Planning & Prep

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**LINKS:** Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.



## Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

### Microbiology Labs



Illustrated Guide to Home Biology Experiments



## Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

(No Subject Supplies Assigned)



## Quick Links

(No Quick Links Assigned)

[Click THIS text  
or scan the QR  
code for links.](#)



# Microbiology Labs

## How To Approach Labs



### Introduce

- Regardless of how many days are required to complete a particular activity, every Science Lab has the same flow, which follows the scientific method.
- Relevant concept(s) are introduced in the text.
- Your notebook entry begins with the introductory/prelab narration, including relevant information that you have read or previously experienced, what you plan to do in the lab, and any hypothesis or anticipated result.



### Lab Procedure

- Perform the procedure according to the instruction, recording in your notebook what you do as it happens. This can be a challenge, but is an extremely important skill.
- Record all data and observations in the lab notebook.



### Analysis & Conclusions

- After all data is collected, analyze the results by considering how the data reflects the introduced concepts and whether the hypothesis is supported by the data.
- If the data and observations do not support the hypothesis, reflect on why and what further testing would be interesting or helpful.

# Microbiology Labs

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## Term 1

### WEEK 1 60m Microbiology Labs - Lesson 1

 Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

Continue reading p.3-51. Then, skim through the rest of the book and plot out a rough plan of which experiments you'd like to explore. Order the needed supplies. Home Scientist has a kit that contains the basic supplies needed. You can add on bacterial cultures and prepared slides, if desired. If you order supplies elsewhere, you can still order the DVD with answers to the review questions in the book and high-resolution images as a separate item. As the introductory material in the book says, if you decide to do labs that require bacterial cultures or prepared slides of certain items that you cannot procure locally, you will need to order those items from Home Scientist ahead of time. They suggest that you could use the high-resolution images that come with their DVD kit instead of purchasing slides.

#### → VIEW

∞ Link: Home Scientist

### WEEK 2 60m Microbiology Labs - Lesson 2

 Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

Today, you will complete LAB I-1: Using a Microscope. Gather all of the needed materials. Prepare your Lab Notebook for the experiment. Read the lab carefully before beginning and adhere to all safety requirements. Record what you do and what you observe in your Lab Notebook. Then clean up and dispose of any materials as instructed.

Complete the lab write-up in your notebook, including the review questions. After you have finished your work, use the answers in the PDF that came with your supplies and see if there was anything significant that you did not understand. Use your Biology textbook from Grade 10 and the internet to learn more or aid in understanding terms and concepts.

You will complete your lab report following the book's guidelines. Choose 6 throughout the year to complete as full lab reports. Use what you have learned about writing lab reports. Use the guidelines for the rest of your reports.

### WEEK 3 60m Microbiology Labs - Lesson 3

 Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 4 60m Microbiology Labs - Lesson 4

 Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 5 60m Microbiology Labs - Lesson 5

 Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

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## Term 1

### WEEK 6 60m Microbiology Labs - Lesson 6

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 7 60m Microbiology Labs - Lesson 7

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 8 60m Microbiology Labs - Lesson 8

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 9 60m Microbiology Labs - Lesson 9

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 10 60m Microbiology Labs - Lesson 10

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 11 60m Microbiology Labs - Lesson 11

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 12 60m Microbiology Labs - Lesson 12

*Exam Week*

→ EXAMS

Answer question(s) related to course.

# Microbiology Labs

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## Term 2

### WEEK 1 60m Microbiology Labs - Lesson 13

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

Continue lab work, and write up your report after completing each lab.

### WEEK 2 60m Microbiology Labs - Lesson 14

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 3 60m Microbiology Labs - Lesson 15

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 4 60m Microbiology Labs - Lesson 16

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 5 60m Microbiology Labs - Lesson 17

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 6 60m Microbiology Labs - Lesson 18

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 7 60m Microbiology Labs - Lesson 19

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 8 60m Microbiology Labs - Lesson 20

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 9 60m Microbiology Labs - Lesson 21

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 10 60m Microbiology Labs - Lesson 22

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

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## Term 2

|                                                                                                                                                                                           |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div>WEEK 11</div> <div><div><div></div></div>60m Microbiology Labs - Lesson 23</div> <div><div>Materials: Illustrated Guide to Home Biology Experiments</div></div> <div>→ LAB DAY</div> |  |
| <div>WEEK 12</div> <div><div><div></div></div>60m Microbiology Labs - Lesson 24</div> <div>Exam Week</div> <div>→ EXAMS</div> <div>Answer question(s) related to course.</div>            |  |



# Microbiology Labs

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## Term 3

### WEEK 1 60m Microbiology Labs - Lesson 25

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

Continue lab work, writing up your report after completing each lab.

### WEEK 2 60m Microbiology Labs - Lesson 26

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 3 60m Microbiology Labs - Lesson 27

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 4 60m Microbiology Labs - Lesson 28

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 5 60m Microbiology Labs - Lesson 29

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 6 60m Microbiology Labs - Lesson 30

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 7 60m Microbiology Labs - Lesson 31

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 8 60m Microbiology Labs - Lesson 32

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 9 60m Microbiology Labs - Lesson 33

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 10 60m Microbiology Labs - Lesson 34

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

# Microbiology Labs

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| Term 3                                                                                                                                                            |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div>WEEK 11</div> <div><div>60m Microbiology Labs - Lesson 35</div><div>Materials: Illustrated Guide to Home Biology Experiments</div><div>→ LAB DAY</div></div> |  |
| <div>WEEK 12</div> <div><div>60m Microbiology Labs - Lesson 36</div><div>Exam Week</div><div>→ EXAMS</div><div>Answer question(s) related to course.</div></div>  |  |